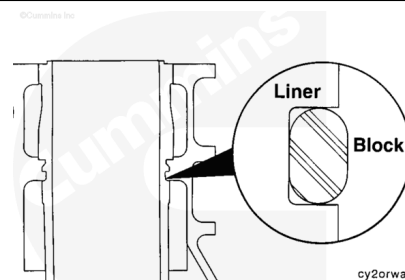


Trainings ▾

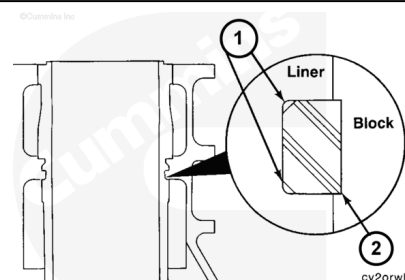
Install

The present o-ring has been produced by two distinctly different manufacturing processes. Following are those two methods and the installation procedures:

The molded o-ring has a symmetrical cross section. The edges are rounded. It does **not** require specific care in orientation of the o-ring other than normal prevention of rolling the o-ring.

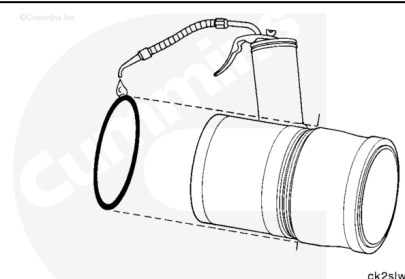


The lathe-cut o-ring has a straight outside with sharp corners. The inside has two 45-degree chamfers (1) so there are no sharp corners next to the liner. This o-ring **must** be installed with the straight side (2) facing the block and the chamfered side next to the liner. If the o-ring is **not** installed this way, liner bore distortion can occur.



⚠ CAUTION ⚠

The liners must be installed within 30 seconds after being lubricated with oil. If the liners are not installed within this time limit, the o-rings will swell and be damaged when the liner is installed.



Make sure the cylinder block and all parts are clean before assembly.

Use clean 15W-40 oil to coat the liner o-ring seals.

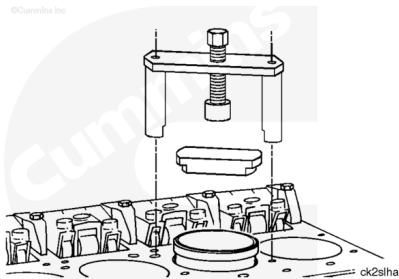
Install the o-ring seal on the liner.

Install the liner into the cylinder block.

When acceptable reused liners are installed, rotate the liners 90 degrees from their original positions in the engine. The thrust and anti-thrust surfaces **must** face the front and back of the cylinder block.

Use cylinder liner driver, Part Number 3824272, and a leather mallet to drive the liner into the cylinder block bore.

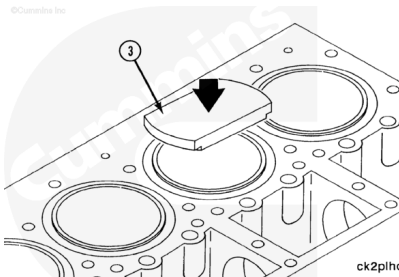
If the liner does **not** seat properly, remove the liner. Inspect the counterbore seat and liner for nicks, burrs, or dirt. Install the liner again.



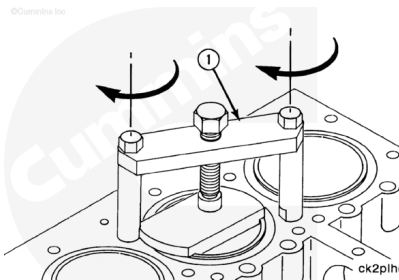
The cylinder liner protrusion **must** be checked before installing the pistons and connecting rods.

Use the cylinder liner clamping tool, Part Number 3824272, to seat the liner in the cylinder block.

Place the force plate (3) across the top of the liner with the step inside the liner bore.



Place the installation tool bridge (1) across the liner and install the two cylinder head capscrews finger-tight to hold the bridge down. Rotate the force plate until the areas where the protrusion measurements will be taken are exposed.

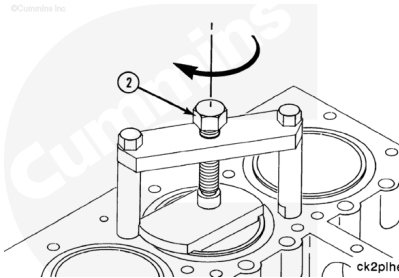


⚠ CAUTION ⚠

Do not use an impact wrench to tighten the liner force plate screw. Excessive torque will cause damage to the liner installation tool.

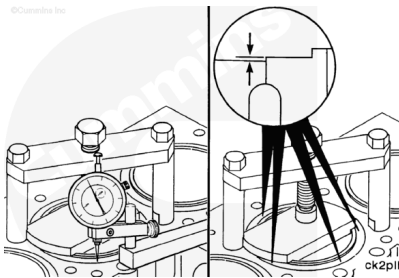
Tighten the forcing screw (2).

Torque Value: 136 n•m [100 ft-lb]



Use a depth gauge, Part Number 3164438, to measure the liner protrusion at four points that are 90 degrees apart.

Cylinder Liner Protrusion		
mm		in
0.00	MIN	0.0000
0.12	MAX	0.0047

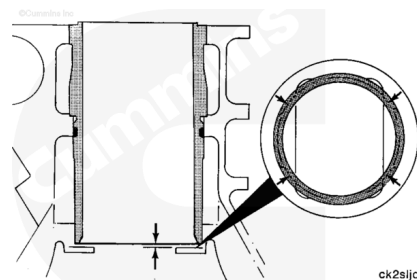


Maximum allowable variation in liner protrusion for each 180 degrees is 0.03mm [0.001 inch].

Use a feeler gauge to measure the liner-to-block clearance at the four-block casting points.

Cylinder Liner-to-Block Clearance

mm		in
0.25	MIN	0.010



Last Modified: 26-Feb-2016